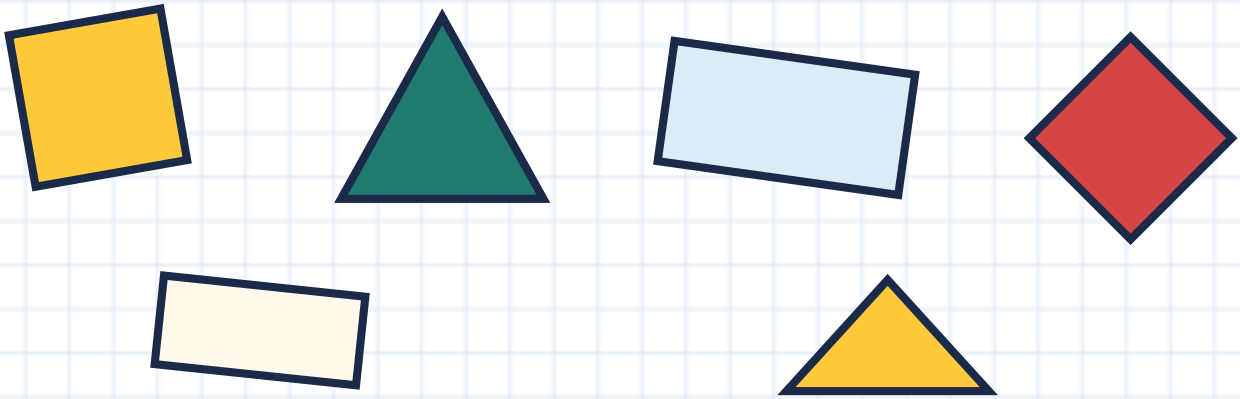


For parents

PRIMARY 1 • MATH INQUIRERS HOLIDAY PROGRAM

PRACTICE AT HOME

Extra questions for every lesson, with answers at the back. **Notice it. Test it. Explain why.**

How to help

No.1 Ask before you tell. Start with "What do you notice?" and let your child answer first.

No.2 Use things you have at home: coins, notes, buttons, spoons, pasta, building blocks. Try it with objects before writing.

No.3 If your child is stuck, give hints in three steps: look again, try it with objects, then try a smaller example. Say the answer last.

No.4 Ask "How do you know?" even when the answer is right. The reason matters as much as the answer.

No.5 Praise the thinking: "I like how you checked that." Avoid "You are clever".

No.6 Keep it short. Ten minutes, two or three questions, is enough. Stop while it is still fun.

No.7 Answers are at the back. Please look only after your child has tried.

The three levels

Core is for everyone. **Think Harder** stretches the idea. **Challenge** is for strong thinkers, and it is fine to leave it for another day. Every question has space to draw or write how your child knows.

COUNT WITH THE FROG

Core

No.1

Write the next two numbers and the rule: 4, 7, 10, 13, ___, ___

No.2

Write the next two numbers and the rule: 5, 10, 15, 20, ___, ___

No.3

Write the next two numbers and the rule: 1, 5, 9, 13, ___, ___

No.4

Write the next two numbers and the rule: 50, 45, 40, 35, ___, ___

Think Harder

No.5

Fill in the missing numbers: 3, 8, ___, 18, ___, 28

No.6

Fill in the missing numbers: 40, ___, 28, ___, 16, 10

No.7

A pattern starts at 3 and adds 5 each time. What is the 8th number?

Challenge (for strong thinkers)

No.8

Frog A jumps in 3s from 0. Frog B jumps in 4s from 0. Which numbers up to 40 do both frogs land on? (Do not count 0.)

No.9

I am in the counting patterns of 3s and of 5s. I am more than 10 and less than 40. Which numbers can I be?

Explain why: how do you know your answer is right?

NUMBER BOND INQUIRY

Core

No.1

10 = 3 + ___

No.2

10 = ___ + 8

No.3

10 = 6 + ___

No.4

20 = 5 + ___

Think Harder

No.5

5 + 4 = 6 + ___

No.6

8 + 2 = 7 + ___

No.7

9 + 6 = ___ + 10

Challenge (for strong thinkers)

No.8

Put 1, 2, 3, 4, 5 and 6 in the triangle so each side adds to 9. Corners 1, 2 and 3. What goes in the middle of each side?

No.9

Corners 4, 5 and 6, and the numbers 1, 2, 3 in the middles. What do all the sides add to? Where does each middle number go?

Explain why: how do you know your answer is right?

LET'S PLAN THE FRUIT PARTY

Core

No.1

A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). Which fruit got the most votes?

No.2

A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). Which fruit got the fewest votes?

No.3

A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many more children chose orange than grape?

No.4

A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many children voted altogether?

Think Harder

No.5

1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children chose blue?

No.6

1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children voted altogether?

No.7

1 picture = 2 children. Red has 3 pictures and blue has 5. How many more children chose blue than red?

Challenge (for strong thinkers)

No.8

18 children voted. 7 chose apple and 5 chose orange. The rest chose grapes. How many chose grapes?

No.9

Votes: A 8, B 5, C 6. Then 4 more children join and all choose B. Which fruit wins now, and by how many votes?

Explain why: how do you know your answer is right?

PATTERN INQUIRY

Core

No.1

The pattern is circle, square, triangle, circle, square, triangle, circle ... What is the 10th one?

No.2

The pattern is star, heart, star, heart, star, heart ... What is the 9th one?

No.3

The pattern is red, red, blue, red, red, blue, red ... What is the 8th one?

No.4

The pattern is A, B, C, D, A, B, C ... What is the 11th one?

Think Harder

No.5

Pattern: circle, triangle, triangle, square, circle, triangle, triangle, square ... What is the 14th shape?

No.6

Same pattern: how many triangles are in the first 20 shapes?

No.7

Same pattern: in which places do the squares come? Write the first five.

Challenge (for strong thinkers)

No.8

Squares in a row use 4 sticks for 1 square, 7 sticks for 2 squares and 10 sticks for 3 squares. How many sticks for 8 squares?

No.9

Using the same pattern, how many squares can you make with 22 sticks?

Explain why: how do you know your answer is right?

MYSTERY NUMBER

Core

No.1

I think of a number. I add 3. I get 8. What is my number?

No.2

I think of a number. I add 6. I get 14. What is my number?

No.3

I think of a number. I add 5. I get 12. What is my number?

No.4

I think of a number. I add 9. I get 17. What is my number?

Think Harder

No.5

I think of a number. I add 5, then I take away 3. I get 9. What is my number?

No.6

I think of a number. I add 6, then I take away 2. I get 10. What is my number?

No.7

I think of a number. I add 4, then I take away 7. I get 8. What is my number?

Challenge (for strong thinkers)

No.8

I think of a number. I add 6, take away 4, then add 5. I get 17. What is my number?

No.9

A number machine has two steps: add 3, then add 4. IN 5 gives OUT 12. What is IN when OUT is 20?

Explain why: how do you know your answer is right?

COIN COMBOS

Core

No.1

Pay exactly 15¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

No.2

Pay exactly 25¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

No.3

Pay exactly $15 with the fewest notes. The notes are $2, $5, $10 and $50.

No.4

Pay exactly $26 with the fewest notes and coins. You have $1 coins and $2, $5, $10 and $50 notes.

Think Harder

No.5

How many ways can you make 25¢ using 20¢, 10¢ and 5¢ coins?

No.6

How many ways can you make 30¢ using 20¢, 10¢ and 5¢ coins?

No.7

How many ways can you make 20¢ using only 10¢ and 5¢ coins?

Challenge (for strong thinkers)

No.8

Make 35¢ using exactly 3 coins from 5¢, 10¢, 20¢ and 50¢. How many ways are there?

No.9

Make 40¢ using exactly 4 coins from 5¢, 10¢, 20¢ and 50¢. Find them all.

Explain why: how do you know your answer is right?

SHAPE SAFARI

Core

No.1

How many sides and corners does a triangle have?

No.2

How many sides and corners does a square have?

No.3

How many sides and corners does a pentagon have?

No.4

How many sides and corners does a hexagon have?

Think Harder

No.5

I have 4 sides. All my sides are the same length. What shape am I?

No.6

How many sides do 3 triangles have altogether?

No.7

How many corners do 5 squares have altogether?

Challenge (for strong thinkers)

No.8

A square has both diagonals drawn. How many triangles can you find?

No.9

A big triangle has 1 line from the top corner to the bottom. How many triangles? What about 2 lines? 3 lines?

Explain why: how do you know your answer is right?

LOGIC LAB

Core

No.1

Ali is taller than Ben. Ben is taller than Mei. Who is the tallest? Who is the shortest?

No.2

The cat is heavier than the hen. The hen is heavier than the mouse. Who is the lightest?

No.3

Red, blue and green ribbons. Red is longer than blue. Green is shorter than blue. Which ribbon is the longest?

No.4

Which number is the odd one out: 2, 4, 6, 9, 10? Why?

Think Harder

No.5

In a race, Sam is ahead of Tia. Tia is ahead of Raj. Mei is ahead of Sam. Who is third?

No.6

Ana, Bo and Cy each have a different toy: a ball, a kite or a yo-yo. Bo does not have the kite. Cy has the ball. Who has which toy?

No.7

Numbers 1 to 12. Which do we say when counting in 2s? Which when counting in 3s? Which in both?

Challenge (for strong thinkers)

No.8

Four friends stand in a line. Ali is not first. Ben is ahead of Ali. Sam is last. Mei is behind Ali. In what order do they stand?

No.9

Three boxes are labelled “Apples”, “Oranges” and “Mixed”. Every label is wrong. You may take one fruit from one box. Which box should you pick from, and why?

Explain why: how do you know your answer is right?

MEASURE IT!

Core

No.1

A pencil is 7 cubes long. A pen is 10 cubes long. How many cubes longer is the pen?

No.2

A ribbon is 12 cubes long. You cut off 5 cubes. How long is the piece left?

No.3

Ali's book is 9 cubes long and Mei's book is 6 cubes long. Whose is longer? By how much?

No.4

Which is longer: 8 cubes or 8 paper clips? Why can we not tell?

Think Harder

No.5

1 clip is as long as 3 cubes. A pen is 12 cubes long. How many clips long is the pen?

No.6

1 clip is as long as 3 cubes. A ruler is 5 clips long. How many cubes long is the ruler?

No.7

Ali's desk is 6 spans long. Mei's desk is 9 spans long, but they measured the same desk. Whose hand is bigger?

Challenge (for strong thinkers)

No.8

A rope is 20 cubes long. Ali cuts it into pieces that are each 4 cubes long. How many pieces does he get? How many cuts does he make?

No.9

A train has 3 carriages. Each is 6 cubes long. There is a gap of 1 cube between carriages. How long is the whole train?

Explain why: how do you know your answer is right?

CHALLENGE DAY

Core

No.1

Count in 3s: 3, 6, 9, ___, ___

No.2

10 = 4 + ___ and 20 = 12 + ___

No.3

Red 4, blue 6, green 3. How many more are blue than green?

No.4

I think of a number. I add 4. I get 9. What is it?

Think Harder

No.5

A pattern starts at 5 and adds 3 each time. What is the 10th number?

No.6

Ali has 12 stickers. He gives 5 to Mei and then gets 3 more. How many does he have now?

No.7

Which three coins from 5¢, 10¢, 20¢ and 50¢ make 45¢?

Challenge (for strong thinkers)

No.8

Two numbers add to 12 and differ by 2. What are they?

No.9

1 triangle uses 3 sticks, 2 triangles use 5 sticks. How many sticks for 9 triangles? How many triangles with 25 sticks?

Explain why: how do you know your answer is right?

ANSWERS

Lesson 1: Count with the Frog

1 16, 19. Add 3.
2 25, 30. Add 5.
3 17, 21. Add 4.
4 30, 25. Take away 5.
5 13 and 23. Add 5.
6 34 and 22. Take away 6.
7 38
8 12, 24 and 36.
9 15 and 30.

Lesson 2: Number Bond Inquiry

1 7
2 2
3 4
4 15
5 3
6 3
7 5
8 Between 1 and 2: 6. Between 2 and 3: 4. Between 3 and 1: 5.
9 Each side adds to 12. Between 4 and 5: 3. Between 5 and 6: 1. Between 6 and 4: 2.

Lesson 3: Let's Plan the Fruit Party

1 Orange
2 Grape
3 5 (9 – 4)
4 25 (7 + 5 + 9 + 4)
5 10
6 24 (12 pictures, each worth 2: count in 2s)
7 4 (2 more pictures, each worth 2: 2, 4)
8 6
9 B wins with 9, which is 1 more than A.

Lesson 4: Pattern Inquiry

1	circle
2	star
3	red
4	C
5	triangle
6	10. The core circle, triangle, triangle, square has 2 triangles. Count the cores in 4s: 4, 8, 12, 16, 20, so there are 5 cores. Count the triangles in 2s: 2, 4, 6, 8, 10.
7	4, 8, 12, 16, 20
8	25. Keep adding 3: 4, 7, 10, 13, 16, 19, 22, 25.
9	7. Keep the table going: 4, 7, 10, 13, 16, 19, 22 sticks. 22 sticks make 7 squares.

Lesson 5: Mystery Number

1	5
2	8
3	7
4	8
5	7
6	6
7	11
8	10 (17 – 5 + 4 – 6)
9	13 (the same as adding 7)

Lesson 6: Coin Combos

1	5¢ + 10¢
2	5¢ + 20¢
3	$10 + $5 (2 notes)
4	$10 + $10 + $5 + $1 (4 pieces)
5	4 ways. (20+5; 10+10+5; 10+5+5+5; 5+5+5+5+5)
6	6 ways.
7	3 ways. (10+10; 10+5+5; 5+5+5+5)
8	1 way: 20¢ + 10¢ + 5¢
9	2 ways: 20¢ + 10¢ + 5¢ + 5¢; 10¢ + 10¢ + 10¢ + 10¢

Lesson 7: Shape Safari

1 3 sides and 3 corners

2 4 sides and 4 corners

3 5 sides and 5 corners

4 6 sides and 6 corners

5 A square

6 9

7 20

8 8 (4 small and 4 made of two small ones)

9 3, 6 and 10 triangles.

Lesson 8: Logic Lab

1 Ali is tallest; Mei is shortest.

2 The mouse

3 Red

4 9. It is the only odd number.

5 Tia (order: Mei, Sam, Tia, Raj).

6 Cy: ball. Bo: yo-yo. Ana: kite.

7 Counting in 2s: 2, 4, 6, 8, 10, 12. Counting in 3s: 3, 6, 9, 12. Both: 6 and 12.

8 Ben, Ali, Mei, Sam.

9 Pick from the box labelled “Mixed”. Its label is wrong, so it holds only one kind of fruit. If you take out an apple, that box is Apples. The box labelled “Oranges” cannot be oranges, and apples are taken, so it is Mixed. The box labelled “Apples” must be Oranges.

Lesson 9: Measure It!

1 3

2 7 cubes

3 Ali’s, by 3 cubes.

4 We cannot tell. The units are different sizes.

5 4 clips

6 15 cubes

7 Ali’s. A bigger span needs fewer spans to cover the same desk.

8 5 pieces and 4 cuts.

9 20 cubes (18 + 2)

Lesson 10: Challenge Day

1	12, 15
2	6 and 8
3	3
4	5
5	32
6	10
7	20¢ + 20¢ + 5¢
8	5 and 7
9	19 sticks for 9; 12 triangles for 25 sticks.